

CorpusGate: Sensitivity–Aware Routing for Secure LLM Inference

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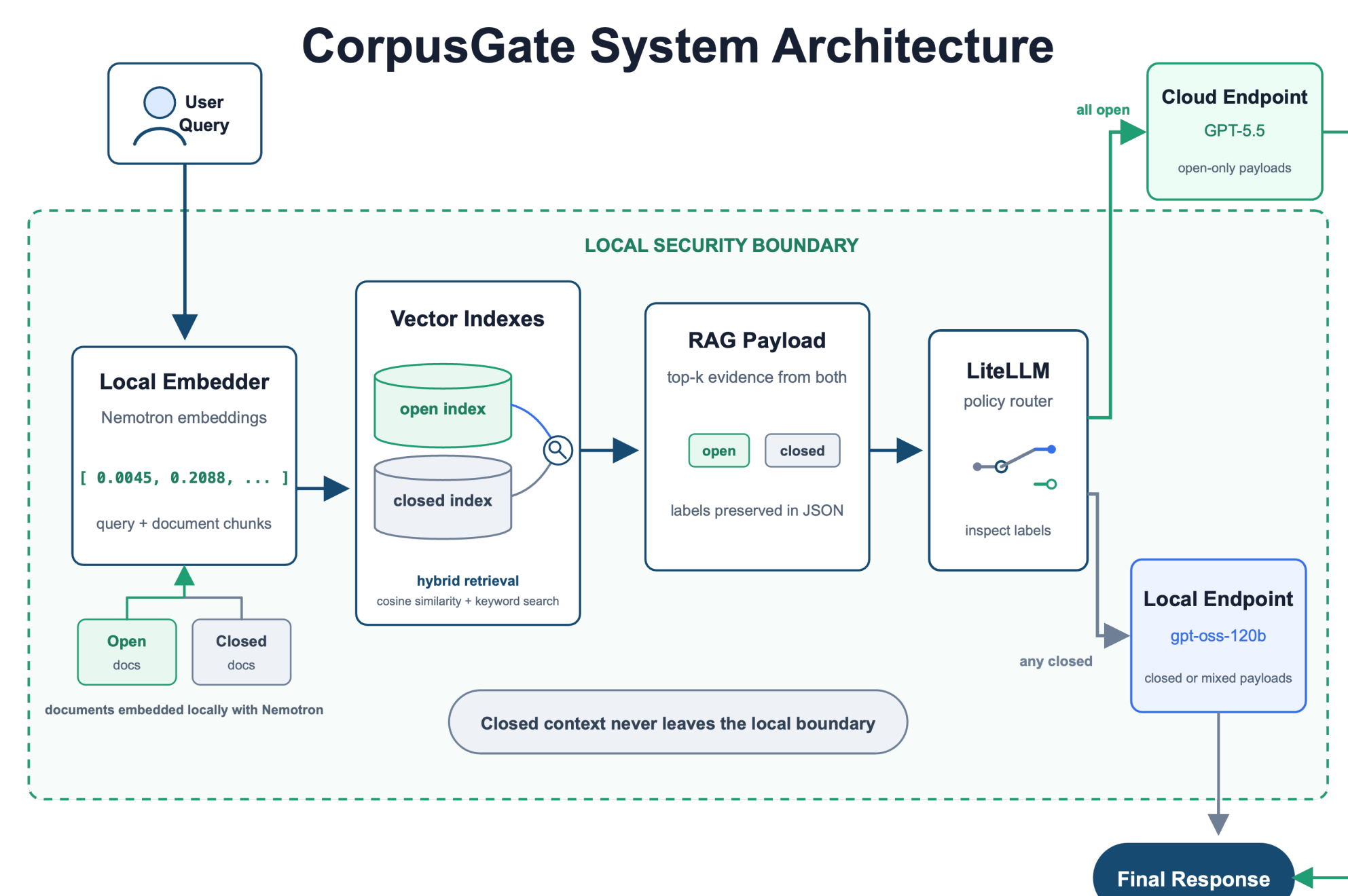
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INTRODUCTION

Many retrieval-augmented generation (RAG) applications operate over document collections that contain both releasable and restricted information. Fully local inference protects sensitive context but often sacrifices the quality gains of larger cloud-hosted models. We implemented CorpusGate, a sensitivity-aware routing system that sends each request to either a local or cloud model based on labels assigned to the retrieved context in the JSON payload. Requests containing only releasable evidence are routed to a cloud model, while any request containing restricted evidence is handled locally. We evaluated CorpusGate on a mixed-sensitivity corpus of DOE project documents using a question-answering benchmark and LLM-as-judge scoring, showing that selective cloud routing can improve answer quality and reduce cloud-token use while keeping restricted evidence inside the local boundary.

OUR APPROACH



- Documents are embedded locally and indexed with preassigned sensitivity labels. For each user query, the system retrieves relevant evidence, preserves each evidence label in the JSON payload, and passes the payload to a LiteLLM [1]-based policy router.
- If all retrieved evidence is releasable, the request is sent to a cloud model; if any retrieved evidence is restricted, the request is routed to a local model.

EVALUATION SETUP & RESULT

- 162 DOE project PDFs were divided into releasable and restricted collections.
- A 240-question testing benchmark was generated using four independent model-and-harness configurations.
- Two blinded LLM-as-judge evaluators (GPT-5.5 and Opus 4.8) scored anonymized answers across five rubric dimensions.

Hybrid Quality Lift Compared to Local

50/50 Releasable/Restricted Document Split		Cloud Model		
Local Model	Model Pairing	gpt-5.5	gpt-5.4-mini	gpt-5.4
	nemotron-3-super-120b	+0.1%	+1.0%	+0.8%
	gemma-4-31b	+1.5%	+1.5%	+1.8%
	gpt-oss-120b	-0.3%	+0.7%	+1.4%
	llama-4-maverick	-0.4%	+0.2%	+0.6%
	llama-3.3-70b	-0.4%	-0.8%	0.0%

80/20 Releasable/Restricted Document Split		Cloud Model		
Local Model	Model Pairing	gpt-5.5	gpt-5.4-mini	gpt-5.4
	nemotron-3-super-120b	+1.8%	+1.7%	+2.2%
	gemma-4-31b	+2.4%	+2.3%	+2.6%
	gpt-oss-120b	+1.6%	+1.7%	+2.1%
	llama-4-maverick	+1.4%	+1.6%	+1.5%
	llama-3.3-70b	+2.2%	+2.3%	+3.3%

*Cell values are percentage lift over the local-only baseline.

Key finding – hybrid routing provides the largest quality gains when most of the document collection is releasable.

Hybrid Quality Decrease Compared to Cloud

50/50 Releasable/Restricted Document Split		Cloud Model		
Local Model	Model Pairing	gpt-5.5	gpt-5.4-mini	gpt-5.4
	nemotron-3-super-120b	-1.7%	-0.2%	-1.7%
	gemma-4-31b	-1.2%	-0.8%	-1.4%
	gpt-oss-120b	-2.0%	-0.8%	-1.3%
	llama-4-maverick	-1.9%	-1.3%	-1.7%
	llama-3.3-70b	-2.4%	-1.9%	-3.6%

80/20 Releasable/Restricted Document Split		Cloud Model		
Local Model	Model Pairing	gpt-5.5	gpt-5.4-mini	gpt-5.4
	nemotron-3-super-120b	-0.1%	0.0%	-0.3%
	gemma-4-31b	-0.3%	-0.1%	-0.6%
	gpt-oss-120b	-0.4%	0.0%	-0.6%
	llama-4-maverick	-0.5%	0.0%	-0.8%
	llama-3.3-70b	-0.2%	0.0%	-0.4%

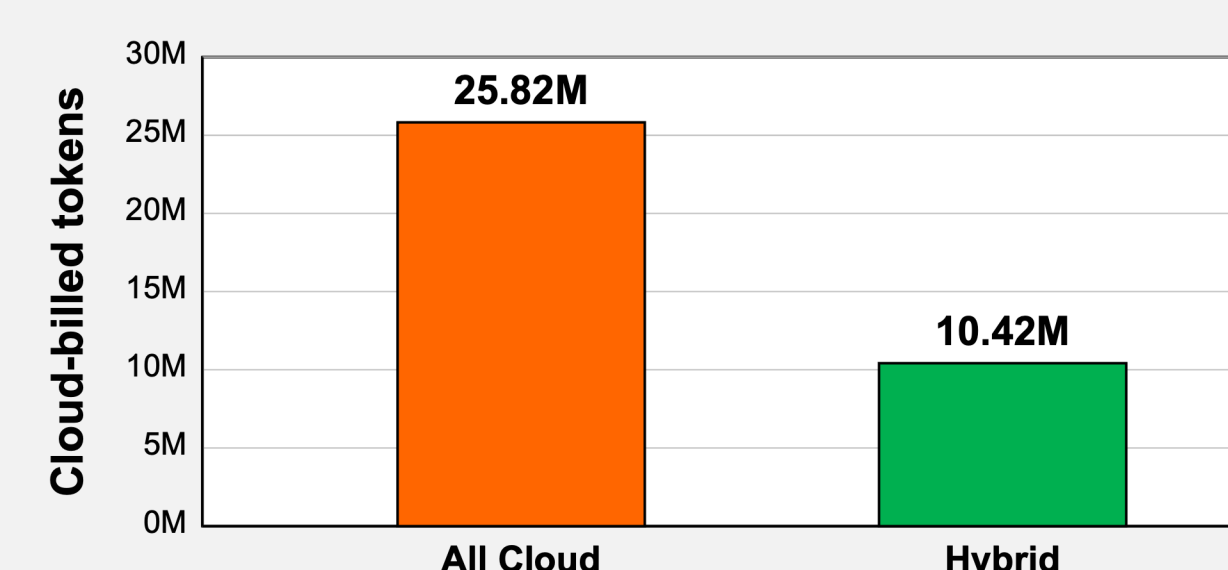
*Cell values are percentage lift over the local-only baseline.

Key finding – in the 80/20 test, hybrid routing closely tracks cloud-only quality while keeping restricted evidence local.

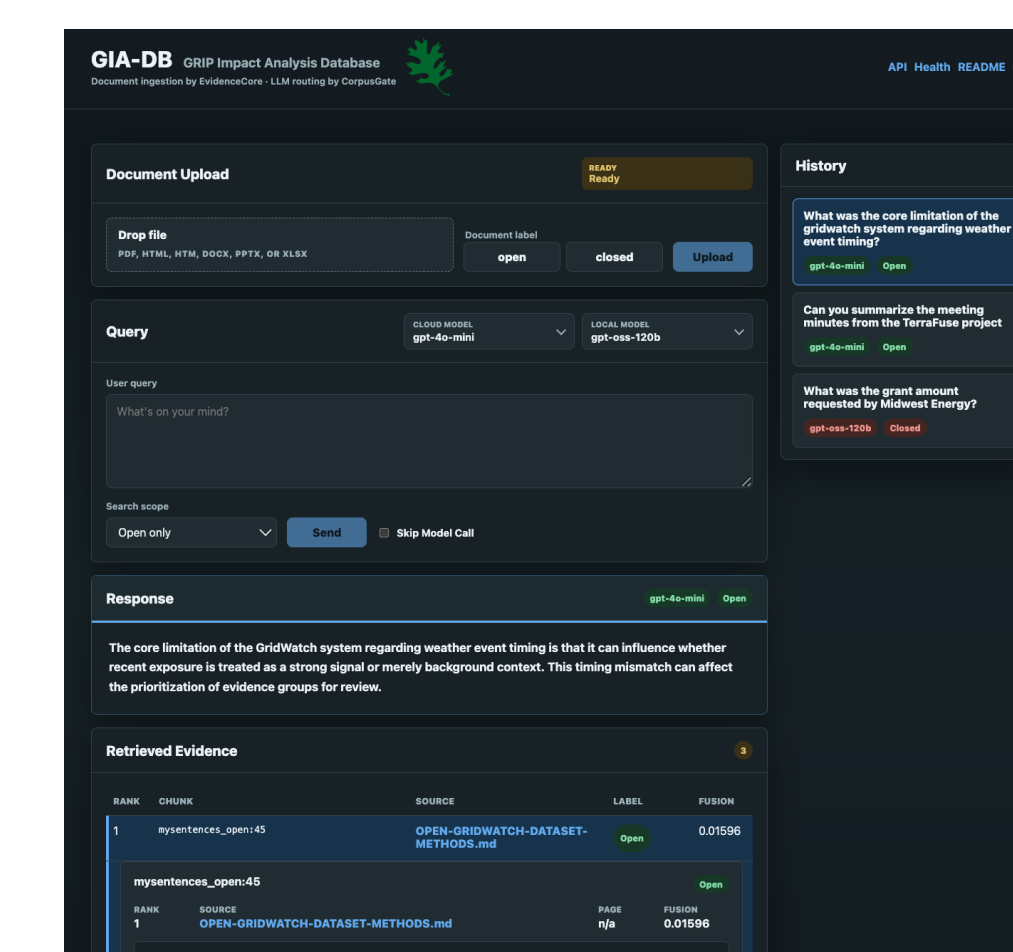
- On the 80/20 releasable/restricted benchmark, CorpusGate outperformed local-only inference by up to **+3.3%**, while closely tracking all-cloud performance.

- CorpusGate avoided **2,235 restricted-context cloud calls** and reduced cloud-billed tokens by **60%** while preserving answer quality.

Cloud-Billed Tokens Across 3,600 Queries



INTEGRATION



CorpusGate prototype user interface

- **GIA-DB** is a secure database supporting impact evaluation of DOE OE's Grid Resilience and Innovation Partnerships (GRIP) program.

- The prototype CorpusGate interface shows how sensitivity-aware routing can be integrated with GIA-DB.

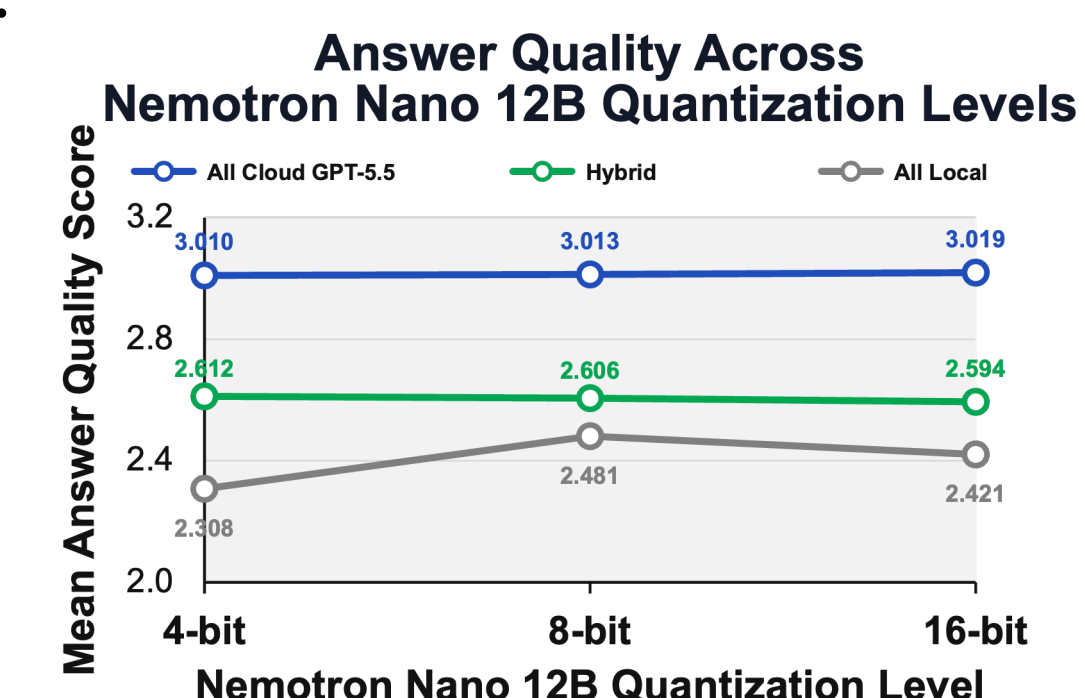
- The interface displays routing decisions, generated responses, and retrieved evidence for each query. It also supports document uploads directly into GIA-DB.

CONCLUSION AND FUTURE WORK

- Restricted retrieved evidence remains inside the local security boundary.
- CorpusGate is most effective when releasable documents make up a larger share of the collection.
- Local model choice matters more as the share of restricted documents increases.
- Hybrid routing reduces cloud-token use while avoiding restricted-context cloud calls.

- Future work will explore the use of a fine-tuned query-sensitivity classifier applied after evidence retrieval to decrease the likelihood of restricted user queries being routed to a cloud-hosted model.

- We benchmarked CorpusGate's answer quality across three quantization levels of Nemotron Nano 12B. Future research will broaden this evaluation to other small local models and quantization variants.



REFERENCES

[1] BerriAI, "LiteLLM AI Gateway (LLM Proxy)," LiteLLM Documentation, 2026. [Online]. Available: https://docs.litellm.ai/docs/simple_proxy. [Accessed: Jun. 29, 2026].